 

# AURORA, IL, USA

WMO: **744655**

Lat: **41.770N** Lon: **88.481W** Elev: **710** StdP: **14.32** Time Zone: **-6.00 (NAC)** Period: **98-19** WBAN: **04808**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-6.0** | **0.2** | **-12.3** | **2.9** | **-5.3** | **-7.2** | **3.8** | **1.2** | **28.3** | **26.8** | **26.0** | **24.0** | **7.6** | **250** | **0.558** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **21.2** | **90.6** | **74.5** | **88.2** | **73.4** | **85.6** | **72.3** | **77.8** | **86.9** | **76.1** | **84.4** | **74.5** | **82.4** | **11.1** | **210** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **75.0** | **134.7** | **83.4** | **73.2** | **126.7** | **81.4** | **72.0** | **121.3** | **79.9** | **41.6** | **86.8** | **39.9** | **84.2** | **38.4** | **82.5** | **84.2** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **26.0** | **23.0** | **19.9** | DB | **-15.6** | **94.3** | **7.1** | **3.2** | **-20.7** | **96.6** | **-24.8** | **98.5** | **-28.8** | **100.3** | **-33.9** | **102.6** | (4) |
| (5) | | | | WB | **-16.0** | **80.8** | **6.8** | **1.5** | **-20.9** | **81.8** | **-24.9** | **82.7** | **-28.7** | **83.5** | **-33.7** | **84.6** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | | DBAvg | **49.3** | **22.7** | **25.3** | **37.6** | **49.1** | **60.2** | **69.6** | **72.8** | **70.5** | **64.1** | **51.7** | **39.2** | **27.8** | (6) |
| (7) | | | DBStd | **19.99** | **12.23** | **12.03** | **10.81** | **9.13** | **8.83** | **6.70** | **5.74** | **5.29** | **7.95** | **9.37** | **9.67** | **11.86** | (7) |
| (8) | | | HDD50 | **3220** | **847** | **694** | **411** | **124** | **13** | **0** | **0** | **0** | **3** | **94** | **342** | **692** | (8) |
| (9) | | | HDD65 | **6470** | **1312** | **1112** | **852** | **481** | **203** | **30** | **6** | **13** | **112** | **422** | **773** | **1154** | (9) |
| (10) | | | CDD50 | **2981** | **0** | **1** | **26** | **97** | **331** | **587** | **708** | **634** | **426** | **148** | **19** | **4** | (10) |
| (11) | | | CDD65 | **757** | **0** | **0** | **2** | **5** | **56** | **167** | **249** | **182** | **85** | **11** | **0** | **0** | (11) |
| (12) | | | CDH74 | **7432** | **0** | **0** | **16** | **86** | **585** | **1643** | **2494** | **1637** | **848** | **123** | **0** | **0** | (12) |
| (13) | | | CDH80 | **2402** | **0** | **0** | **1** | **12** | **170** | **573** | **889** | **476** | **257** | **24** | **0** | **0** | (13) |
| (14) | **Wind** | | WSAvg | **9.1** | **10.6** | **10.8** | **10.7** | **11.4** | **10.0** | **8.2** | **6.5** | **5.8** | **6.7** | **8.5** | **9.8** | **10.1** | (14) |
| (15) | **Precipitation** | | PrecAvg | **35.6** | **1.5** | **1.5** | **2.3** | **3.4** | **4.3** | **4.0** | **3.9** | **4.1** | **3.2** | **2.8** | **2.4** | **2.0** | (15) |
| (16) | | | PrecMax | **45.7** | **3.5** | **3.0** | **4.7** | **7.1** | **7.1** | **8.5** | **6.6** | **10.0** | **13.7** | **7.0** | **5.4** | **4.9** | (16) |
| (17) | | | PrecMin | **21.9** | **0.5** | **0.2** | **0.8** | **1.4** | **1.6** | **0.8** | **1.9** | **1.8** | **0.5** | **0.6** | **0.7** | **0.7** | (17) |
| (18) | | | PrecStd | **6.1** | **0.9** | **0.8** | **1.1** | **1.4** | **1.5** | **2.0** | **1.6** | **1.9** | **3.1** | **1.6** | **1.2** | **1.2** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | | **55.1** | **62.3** | **76.1** | **81.6** | **89.3** | **92.7** | **95.3** | **91.2** | **90.4** | **83.5** | **69.9** | **60.8** | (19) |
| (20) | | | MCWB | | **51.0** | **54.7** | **63.8** | **63.3** | **69.7** | **73.6** | **77.5** | **75.7** | **74.0** | **69.8** | **58.2** | **57.6** | (20) |
| (21) | | 2% | DB | | **47.3** | **53.2** | **65.8** | **75.8** | **84.4** | **90.0** | **90.8** | **88.1** | **87.0** | **77.2** | **63.5** | **54.1** | (21) |
| (22) | | | MCWB | | **44.4** | **48.1** | **56.7** | **60.8** | **67.4** | **72.6** | **76.2** | **74.6** | **72.0** | **63.7** | **55.4** | **50.2** | (22) |
| (23) | | 5% | DB | | **42.0** | **45.5** | **60.6** | **71.7** | **81.0** | **86.7** | **88.0** | **84.9** | **82.3** | **72.6** | **59.3** | **47.7** | (23) |
| (24) | | | MCWB | | **39.4** | **41.6** | **53.1** | **58.1** | **65.9** | **71.5** | **74.9** | **73.0** | **68.7** | **62.0** | **53.5** | **45.1** | (24) |
| (25) | | 10% | DB | | **37.9** | **40.0** | **54.5** | **66.2** | **76.2** | **82.8** | **84.8** | **82.2** | **79.1** | **67.8** | **54.7** | **43.3** | (25) |
| (26) | | | MCWB | | **35.8** | **37.0** | **48.0** | **55.4** | **64.3** | **69.7** | **72.9** | **71.1** | **67.3** | **59.8** | **49.1** | **40.0** | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | | **52.8** | **56.3** | **63.9** | **66.8** | **73.7** | **78.1** | **80.7** | **78.7** | **76.7** | **71.6** | **61.3** | **58.0** | (27) |
| (28) | | | MCDB | | **54.2** | **59.8** | **74.8** | **77.4** | **84.5** | **87.5** | **89.8** | **87.9** | **86.6** | **81.4** | **66.0** | **60.2** | (28) |
| (29) | | 2% | WB | | **44.6** | **48.5** | **58.8** | **62.9** | **70.5** | **75.4** | **78.6** | **76.6** | **74.1** | **67.2** | **57.4** | **50.7** | (29) |
| (30) | | | MCDB | | **47.3** | **52.8** | **64.8** | **71.9** | **80.2** | **84.4** | **87.6** | **84.9** | **82.8** | **72.2** | **61.6** | **52.7** | (30) |
| (31) | | 5% | WB | | **39.3** | **42.1** | **54.3** | **60.0** | **68.4** | **73.6** | **76.9** | **74.9** | **71.7** | **64.1** | **53.6** | **45.1** | (31) |
| (32) | | | MCDB | | **41.4** | **45.8** | **59.0** | **69.4** | **77.1** | **83.0** | **85.5** | **82.3** | **78.8** | **70.0** | **58.1** | **47.3** | (32) |
| (33) | | 10% | WB | | **36.0** | **37.3** | **48.3** | **56.8** | **66.0** | **71.7** | **75.1** | **73.1** | **69.4** | **60.7** | **49.9** | **40.3** | (33) |
| (34) | | | MCDB | | **38.1** | **40.1** | **54.1** | **65.2** | **73.9** | **80.7** | **82.7** | **79.9** | **76.3** | **66.8** | **55.0** | **43.2** | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | | **15.3** | **16.7** | **18.5** | **21.9** | **21.5** | **21.0** | **21.2** | **21.2** | **23.4** | **21.3** | **18.0** | **14.8** | (35) |
| (36) | | 5% DB | MCDBR | | **17.5** | **21.8** | **25.6** | **29.7** | **26.0** | **24.4** | **23.1** | **23.4** | **25.7** | **27.1** | **24.8** | **18.1** | (36) |
| (37) | | | MCWBR | | **14.9** | **17.1** | **17.4** | **17.5** | **12.9** | **11.3** | **11.3** | **12.1** | **12.5** | **15.4** | **17.8** | **15.7** | (37) |
| (38) | | 5% WB | MCDBR | | **16.6** | **21.0** | **22.8** | **25.6** | **22.0** | **20.6** | **20.1** | **19.8** | **21.3** | **22.1** | **21.5** | **16.8** | (38) |
| (39) | | | MCWBR | | **14.6** | **17.2** | **16.9** | **16.7** | **12.5** | **11.3** | **11.3** | **11.4** | **11.4** | **15.0** | **17.1** | **15.3** | (39) |
| (40) | **Clear-Sky Solar Irradiance** | taub | | | **0.291** | **0.308** | **0.341** | **0.395** | **0.422** | **0.430** | **0.443** | **0.432** | **0.405** | **0.351** | **0.308** | **0.291** | (40) |
| (41) | | taud | | | **2.419** | **2.379** | **2.389** | **2.251** | **2.228** | **2.247** | **2.206** | **2.231** | **2.280** | **2.430** | **2.500** | **2.443** | (41) |
| (42) | | Ebn at Noon | | | **273** | **285** | **287** | **277** | **271** | **269** | **264** | **263** | **262** | **266** | **266** | **263** | (42) |
| (43) | | Edh at Noon | | | **25** | **30** | **34** | **42** | **44** | **43** | **45** | **42** | **37** | **29** | **23** | **22** | (43) |
| (44) | **All-Sky Solar Radiation** | RadAvg | | | **523** | **813** | **1142** | **1476** | **1689** | **1920** | **1922** | **1692** | **1416** | **931** | **599** | **440** | (44) |
| (45) | | RadStd | | | **57** | **105** | **92** | **120** | **133** | **139** | **106** | **98** | **121** | **107** | **53** | **47** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD50 | HDD65 | CDD50 | CDD65 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (58 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page